Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Practise-1

2520030078

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Section – 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Output:

```
143  09:12:06  CREATE DATABASE IF NOT EXISTS bookflow5_db
```

Use database:

```
USE bookflow5_db;
```

Output:

```
145  09:14:03  USE bookflow5_db
```

Create Physician Table:

```
CREATE TABLE physician (
    employeeid INT PRIMARY KEY,
    name VARCHAR(100),
    position VARCHAR(100),
    ssn VARCHAR(20)
);
```

Output:

```
146  09:15:11  CREATE TABLE physician (    employeeid INT PRIMARY KEY,    name VARCHAR(100),    position VAR
```

Insert Data Into Physician:

```
INSERT INTO physician VALUES
(1,'Sushma','Staff Internist','111111111'),
(2,'Sowmya','Attending Physician','222222222'),
(3,'Mokshagna','Surgical Attending Physician','333333333'),
(4,'Akhila','Senior Attending Physician','444444444'),
(5,'Likhitha','Head Chief of Medicine','555555555'),
(6,'Vaishnavi','Surgical Attending Physician','666666666'),
(7,'Manasvi','Surgical Attending Physician','777777777'),
(8,'Sairam','MD Resident','888888888'),
(9,'Krishna','Attending Psychiatrist','999999999');
```

Output:

```
147  09:20:16  INSERT INTO physician VALUES (1,'Sushma','Staff Internist','111111111'), (2,'Sowmya','Attending Physician',
```

Select:

```
SELECT * FROM physician;
```

Output:

employeeid	name	position	ssn
1	Sushma	Staff Internist	111111111
2	Sowmya	Attending Physician	222222222
3	Mokshagna	Surgical Attending Physician	333333333
4	Akhila	Senior Attending Physician	444444444
5	Likhitha	Head Chief of Medicine	555555555
6	Vaishnavi	Surgical Attending Physician	666666666
7	Manasvi	Surgical Attending Physician	777777777
8	Sairam	MD Resident	888888888
9	Krishna	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Create Department Table:

```
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

Output:

```
149  09:23:25  CREATE TABLE department (  departmentid INT PRIMARY KEY,  name VARCHAR(100),  head INT )
```

Insert Data Into Department:

```
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
SELECT * FROM department;
```

Output:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

Inner Join:

```
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

Output:

Department	Head_Physician
General Medicine	Akhila
Surgery	Manasvi
Psychiatry	Krishna

Where Clause:

```
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
```

Output:

employeeid	name	position	ssn
3	Mokshagna	Surgical Attending Physician	333333333
6	Vaishnavi	Surgical Attending Physician	666666666
7	Manasvi	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

Where + Like:

```
SELECT *
FROM physician
WHERE name LIKE 'Sushma%';
```

Output:

employeeid	name	position	ssn
1	Sushma	Staff Internist	111111111
NULL	NULL	NULL	NULL

Count():

```
SELECT COUNT(*) AS Total_Physicians
FROM physician;
```

Output:

Total_Physicians
9

Count Distinct:

```
SELECT COUNT(DISTINCT position)
FROM physician
```

Output:

COUNT(DISTINCT position)
7

Group By:

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

Output:

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Group By + Having:

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

Output:

position	Total_Employees
Surgical Attending Physician	3

Order By:

```
SELECT *
FROM physician
ORDER BY name ASC;
```

Output:

employeeid	name	position	ssn
4	Akhila	Senior Attending Physician	444444444
9	Krishna	Attending Psychiatrist	999999999
5	Likhitha	Head Chief of Medicine	555555555
7	Manasvi	Surgical Attending Physician	777777777
3	Mokshagna	Surgical Attending Physician	333333333
8	Sairam	MD Resident	888888888
2	Sowmya	Attending Physician	222222222
1	Sushma	Staff Internist	111111111
6	Vaishnavi	Surgical Attending Physician	666666666
NULL	NULL	NULL	NULL

Order By Desc:

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

Output:

employeeid	name	position	ssn
9	Krishna	Attending Psychiatrist	999999999
8	Sairam	MD Resident	888888888
7	Manasvi	Surgical Attending Physician	777777777
6	Vaishnavi	Surgical Attending Physician	666666666
5	Likhitha	Head Chief of Medicine	555555555
4	Akhila	Senior Attending Physician	444444444
3	Mokshagna	Surgical Attending Physician	333333333
2	Sowmya	Attending Physician	222222222
1	Sushma	Staff Internist	111111111
NULL	NULL	NULL	NULL

Where + In:

```
SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);
```

Output:

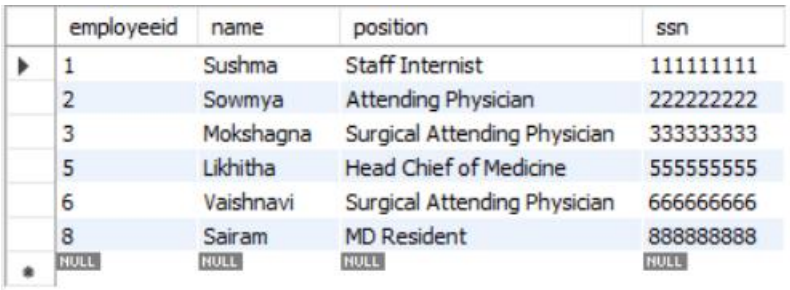


	employeeid	name	position	ssn
▶	4	Akhila	Senior Attending Physician	444444444
	7	Manasvi	Surgical Attending Physician	777777777
	9	Krishna	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

Subquery:

```
SELECT *
FROM physician
WHERE employeeid NOT IN
(
    SELECT head
    FROM department
);
```

Output:



	employeeid	name	position	ssn
▶	1	Sushma	Staff Internist	111111111
	2	Sowmya	Attending Physician	222222222
	3	Mokshagna	Surgical Attending Physician	333333333
	5	Likhitha	Head Chief of Medicine	555555555
	6	Vaishnavi	Surgical Attending Physician	666666666
	8	Sairam	MD Resident	888888888
*	NULL	NULL	NULL	NULL

Group By + Order By:

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
ORDER BY Total_Employees DESC;
```

Output:

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Where + Like

```
SELECT *
FROM physician
WHERE position LIKE '%Attending%';
```

Output:

employeeid	name	position	ssn
2	Sowmya	Attending Physician	222222222
3	Mokshagna	Surgical Attending Physician	333333333
4	Akhila	Senior Attending Physician	444444444
6	Vaishnavi	Surgical Attending Physician	666666666
7	Manasvi	Surgical Attending Physician	777777777
9	Krishna	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Join + Where

```
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```

Output:

name
Manasvi

Join + Order By

```
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

Output:

	name	name
▶	General Medicine	Akhila
	Psychiatry	Krishna
	Surgery	Manasvi

Where + Not Like

```
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
```

Output:

	employeeid	name	position	ssn
▶	1	Sushma	Staff Internist	111111111
	2	Sowmya	Attending Physician	222222222
	4	Akhila	Senior Attending Physician	444444444
	5	Likhitha	Head Chief of Medicine	555555555
	8	Sairam	MD Resident	888888888
	9	Krishna	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

Count + Having

```
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Output:

	position	Total
▶	Surgical Attending Physician	3

Exists

```
SELECT *
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.hea
);
```

Output:

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
*	NULL	NULL	NULL

Max()

```
SELECT MAX(employeeid)
FROM physician;
```

Output:

	MAX(employeeid)
▶	9

Min():

```
SELECT MIN(employeeid)
FROM physician;
```

Output:

	MIN(employeeid)
▶	1